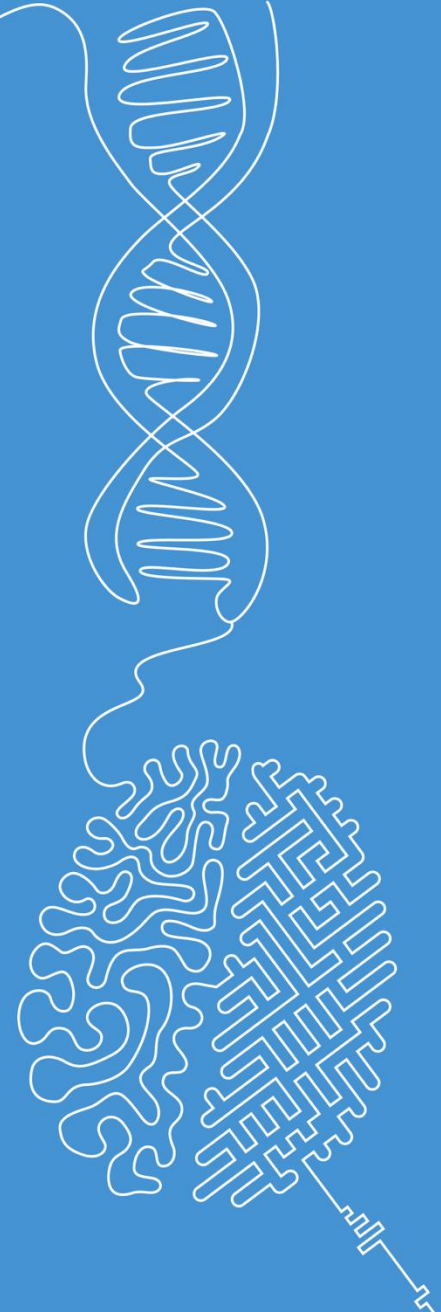


From Time Series to Images

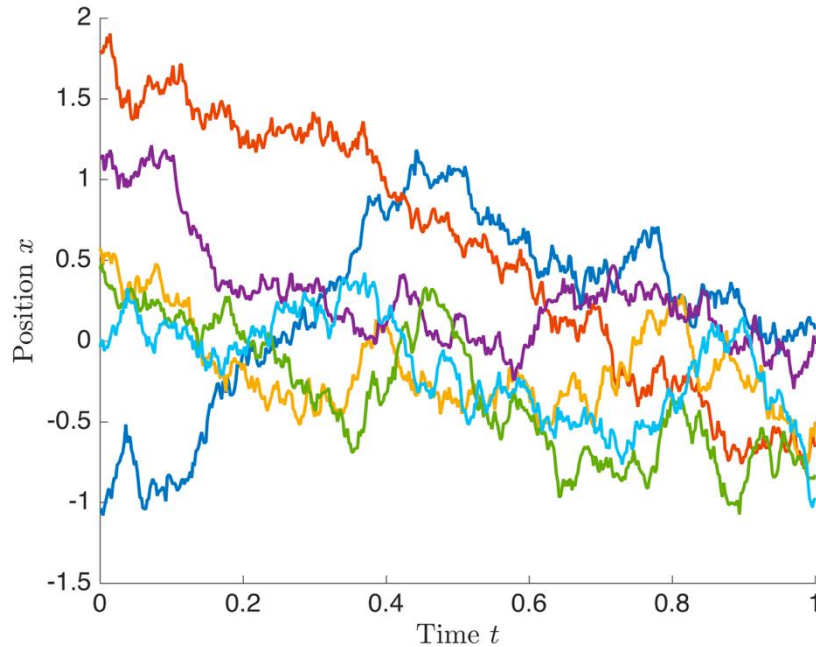
Image and Signal Processing

Norman Juchler



From time series to images...

Time series



Image

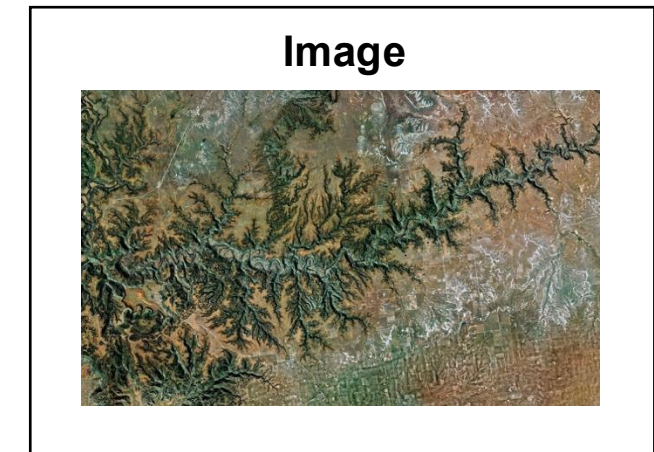
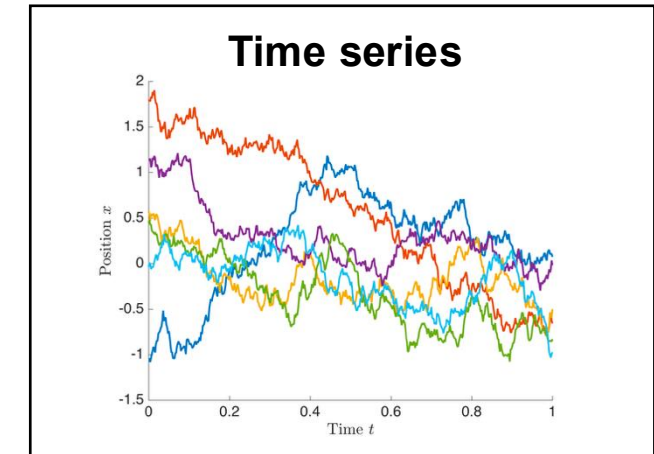


Task: What have these things in common?
What are the differences?

From time series to images...

What they have **in common**:

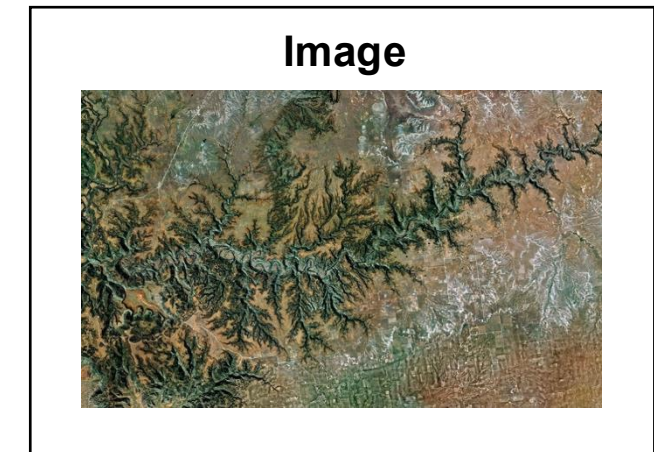
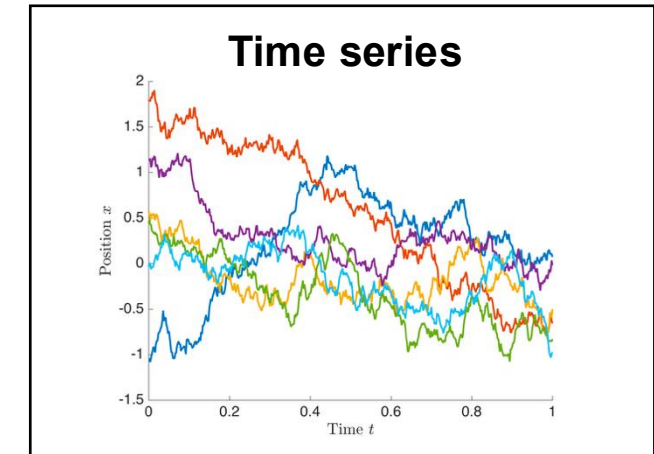
- Both are signals:
 - Measured data that vary over a domain
 - Contain information or patterns
 - May be affected by noise
 - Can be continuous or discrete
- Both can be processed using similar techniques:
 - Filtering (low-pass, high-pass, etc.)
 - Fourier transform
 - Feature extraction
- Local structure matters: Nearby values are related
 - In time-series: neighboring time points are often correlated
 - In images: neighboring pixels are often similar



From time series to images...

What the **differences** are:

Feature	Time Series	Image
Signal dimension	1D signal varying over time: $x(t)$	2D signal varying over space: $I(x,y)$
Axis meaning	Time (temporal axis)	Spatial position (x, y)
Sampling	Typically uniform sampling in time	Discrete grid sampling (pixels) over space
Directionality	Inherently directional (past $\rightarrow$ future, causality)	No inherent directionality; spatially symmetric
Types of patterns	Periodicity, trends, temporal events	Edges, textures, shapes, regions
Visualization	Line plot (amplitude vs. time)	Image display (intensity or color map)



■ Domain shift: **From time domain** $\rightarrow$ **spatial domain**

- We move from analyzing how signals change over time to how they change across space

Topics

- **Discretization in 2D**
 - Sampling
 - Quantization
- **Color spaces**
 - Binary / RGB / RGBA / HSL / CMYK ...
 - Conversions
- **Basic image operations**
 - Cropping
 - Spatial image transformations
- **Spatial Fourier analysis**
 - (Spectrograms in 2D)
 - FFT for images
- **Convolution / filtering / kernels**
 - Noise removal / smoothing
 - Gradients and image derivatives
- **Features**
 - Edge detection
 - Corners / interest points
 - Invariance properties
- **Masks / morphological operations**
 - Thresholding
 - Contours
 - Erosion / dilation / closure / opening

We will continue working with Python...

- OpenCV
- Pillow/PIL
- scikit-image
- Numpy / SciPy / Matplotlib
- Mahotas
- ITK/SimpleITK



Resources

Books

- *Computer Vision: Algorithms and Applications*. Szeliski, 2021: [Link](#)
- *Digital Image Processing*. Gonzalez and Woods, 2009: [Link](#)
- *Programming Computer Vision with Python*. Jan Erik Solem, 2012: [Link](#)

Online course:

- *First principles of computer vision*. Nayar, 2022. [Link](#)

Tutorials

- Image processing (with Pillow): [Real Python](#)
- Image processing (with OpenCV): [OpenCV.org](#)
- Digital image processing tutorial: [tutorialspoint](#)